Pasquini Giuseppe

Arco, Trento, Italy

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At the moment, I'm enrolled in the Master's degree in Mechatronics Engineering at the University of Trento. My passion lies in the fields of mechatronics, robotics, and automation, with a commitment to sustainability. During my academic journey, I have engaged in diverse scientific projects, reinforcing my belief in the power of collaborative teamwork to overcome challenges effectively.

Education

MSc Mechatronics Engineering | University of Trento — **Sept 2022 - Dec 2025**
Thesis: Self-powered co-harvesting for batteryless sensing in two-wheeled IoT applications — *Trento, Italy*

BSc Computer, Communication and Electronic Engineering | University of Trento — **Sept 2019 - Sept 2022**
Thesis: Streamlined IT processes through lean philosophy (Italian) — *Trento, Italy*

Publications

Self-powered co-harvesting for batteryless cycling IoT applications | *IEEE STAR 2025* — **Oct 2025**
Trento, Italy
The paper presents a hybrid energy-harvesting system for downhill bicycles that addresses the problem of multiple onboard batteries by recovering energy from the ride. The solution enables batteryless telemetry and self-powered actuators, without affecting downhill performance.

Experience

IT Engineering Intern | *ZF Group - Business Unit Marine & Special Driveline Technology* — **Feb 2022 - Jun 2022**
Deductive reasoning, problem solving, system configuration, documentation — *Arco & Padova, Italy*
- Development of procedures for the resolution of recurring IT issues/activities
- Verification and drafting of technical infrastructure documentation, as well as management of IT databases and devices
- Weekly daily trips to the second facility in Selvazzano Dentro (PD) - **Reference letter available**

Electrical Engineering Intern | *Ace Control System Limited* — **Aug 2018 - Sept 2018**
Communication, teamwork, electrical wiring, electrical controls — *Cork, Ireland*
- Assistance during design and wiring of electrotechnical systems/control panels (civil and industrial applications)
- Fault analysis and testing of electrotechnical systems

Research And Development - Testing Intern | *ZF Group - Marine Business Unit* — **May 2017 - Jun 2017**
Problem solving, knowledge acquisition, teamwork, R&D — *Arco, Italy*
- Study, creation, and testing of mechatronics systems
- Embedded systems failure analysis and assistance during transmission testing

Rifugio Val di Fumo — **Summer Seasons 2019 - 2022**
- Employed during the summer seasons, worked in the tourism sector — *Valdaone, Italy*

Projects

Real-Time Bike Monitoring via Sustainable IoT and Android App | *Grafana, InfluxDB, C++* — **Oct 2024**
- Use Arduino Nano, Raspberry Pi, and smartphone GPS to capture and visualize data, displayed on a Grafana dashboard.

Optimal Control with Actor-Critic Learning | *Python* — **Sept 2024**
- Developed Actor-Critic learning via neural network training for value function approximation and policy refinement.

A Simulation of an Advanced Autonomous Submarine Navigation | *Matlab* — **Jul 2024**
- Submarines navigation with Kalman filtering for leader positioning, consensus control for follower formation.

Stabilization Control of a Multirotor UAV | *Matlab* — **Jul 2023**
- Optimal controller design for asymptotic stabilization of a simplified dynamics drone model at a target height.

Walking Assistant Robot | Mechanical Design | *Structural analysis* — **Feb 2023**
- Design of mechanical system (walking assistant robot) subject to static and dynamic loads and fatigue analysis.

High Precision Linear Displacement Measurement System | *LTspice, Matlab* — **Jan 2023**
- Design of a measurement system based on light's duty cycle evaluation, with a resolution in the order of nanometers.

Design and Realization of a LAN-NTP embedded system | *KiCAD, C, Visual Studio Code* — **Jun 2022**
- Designed, mounted, and programmed an NTP system to correct RTC drift using GPS, accessible via Ethernet.

Technical Skills

Certifications: Comau robotics: use and programming for the C5G family, ECDL Full standard
Languages: Python, C, C++, Java, SQL
Engineering Tools: KiCAD, LTspice, Matlab (Simulink), Ansys, Maple, Mathematica, AutoCAD